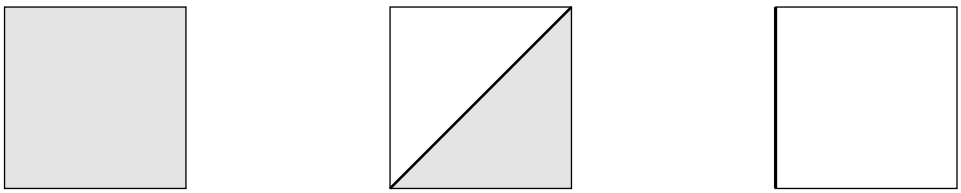
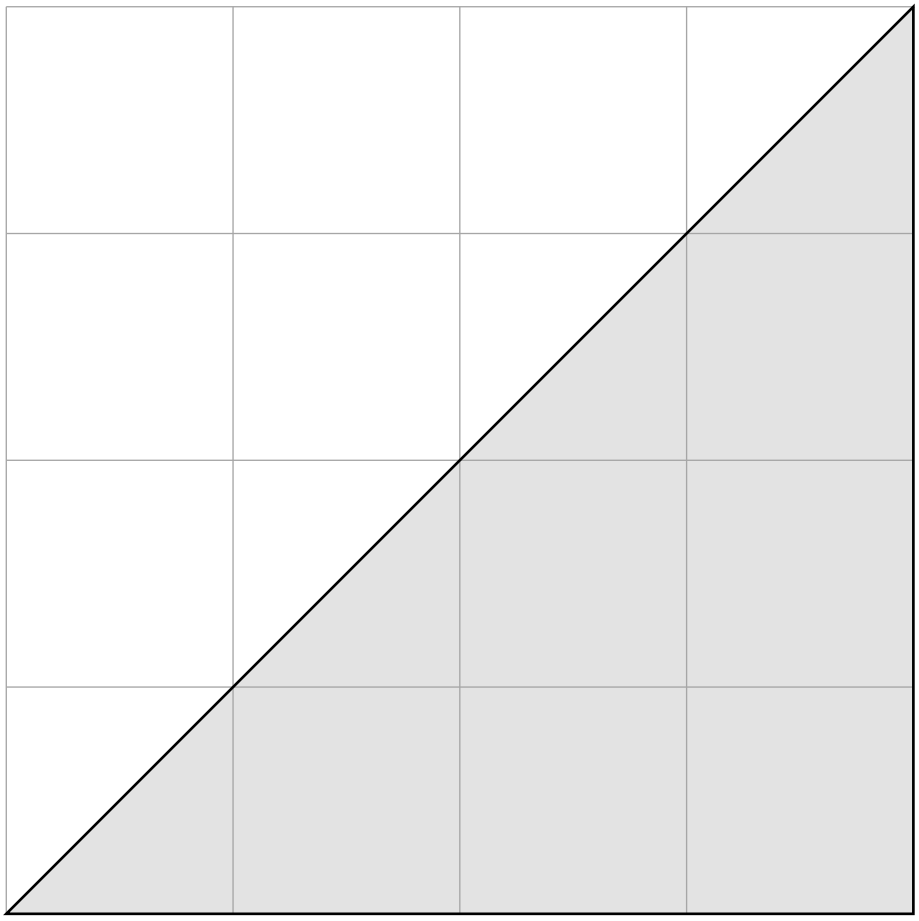
Use whole squares inside the gray shape. Cover all squares with some gray area. A touch at an edge or corner adds no square.



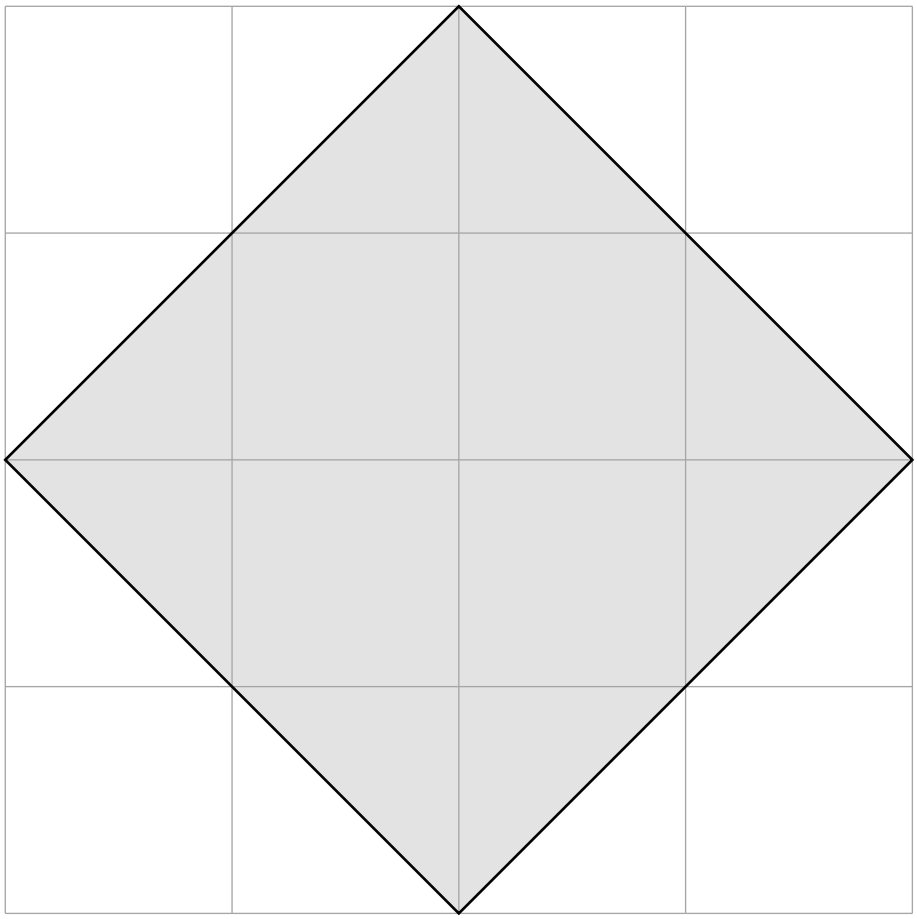
whole partial outside

Problem 1: Find an inside cover and an outside cover for the triangle. Use as many whole inside squares and as few outside-cover squares as possible.



Inside: _____ big squares Cover: _____ big squares

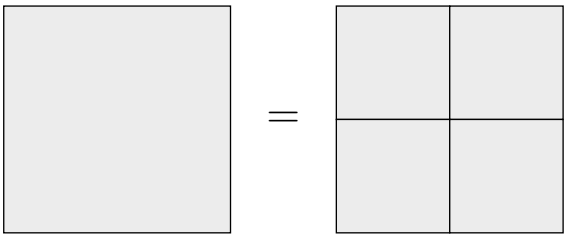
Problem 2: Trap the diamond's area between an inside cover and an outside cover. Can you be certain its area is more than 3 big squares? Less than 13?



Inside: _____ big-square units

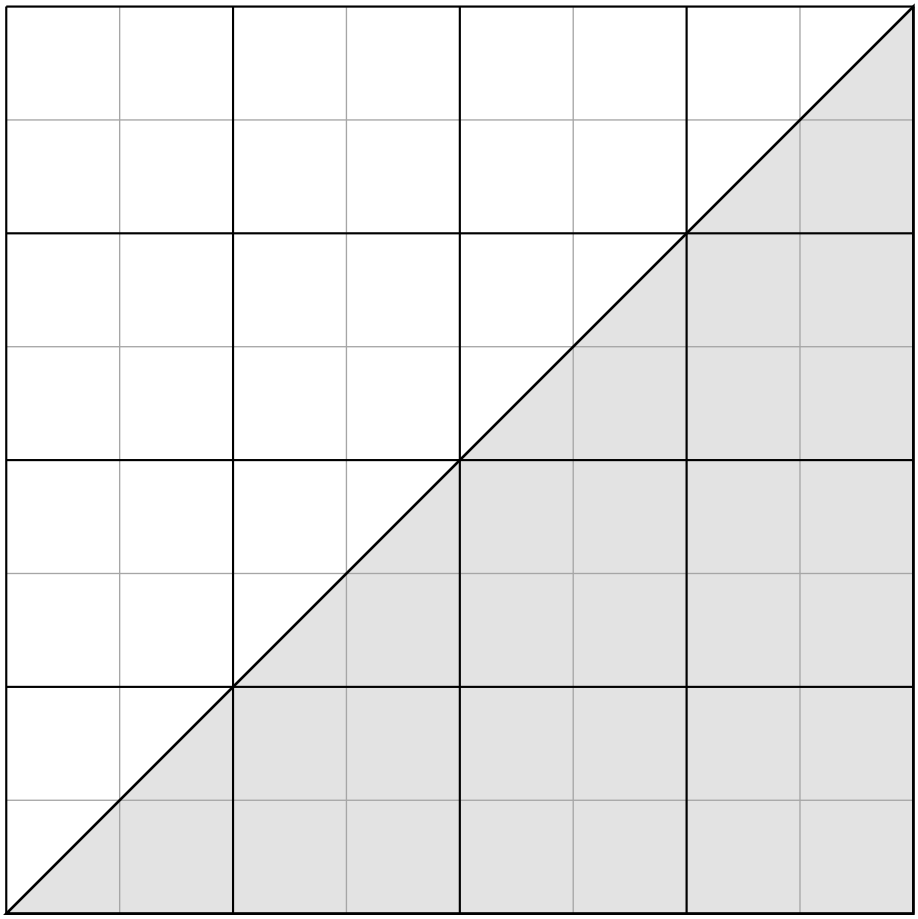
Cover: _____ big-square units

One grid square is 1 big-square unit.



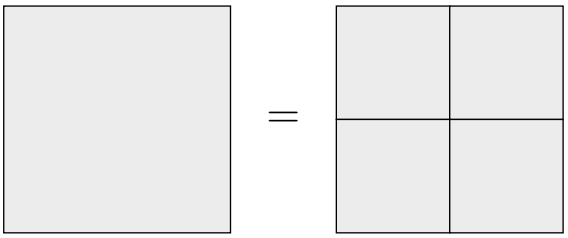
1 big square
= 4 small squares

Problem 3: Use the smaller squares to trap the triangle's area more tightly. State both bounds in big-square units.



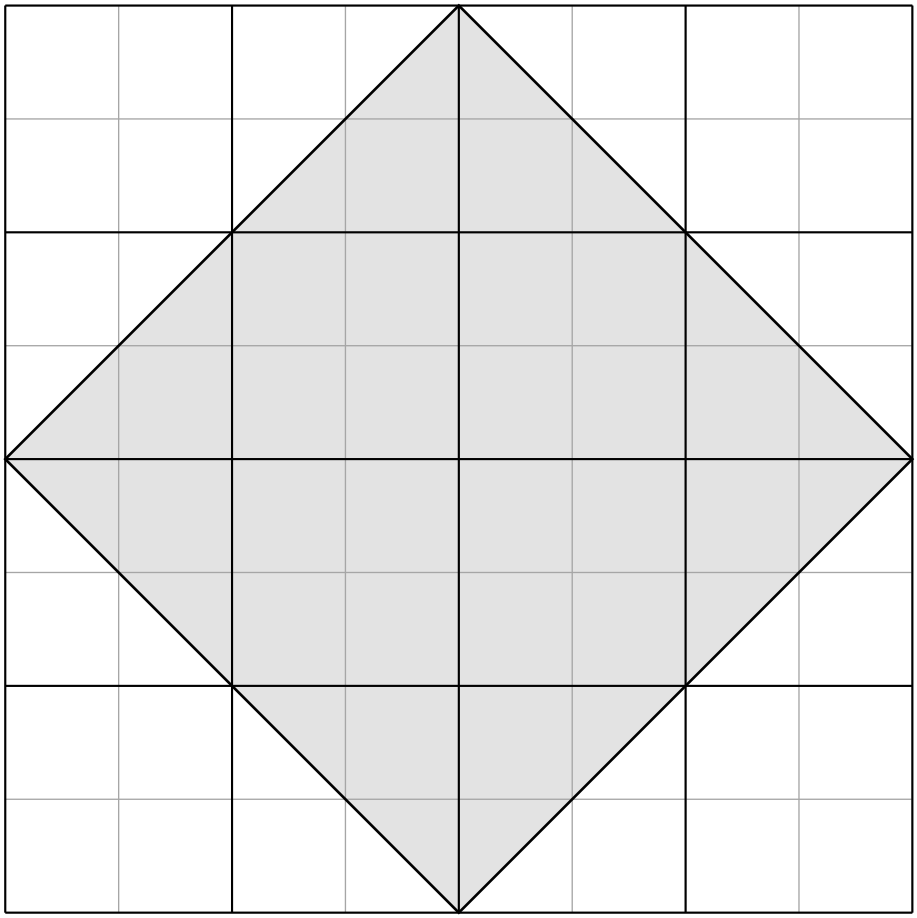
Inside: _____ big-square units

Cover: _____ big-square units



1 big square
= 4 small squares

Problem 4: Find new bounds for the diamond. Compare the area left uncertain on the big grid and on the small grid.



Inside: _____ big-square units

Cover: _____ big-square units